```
> restart:
  with(VectorCalculus):
  with(linalg):
  with(LinearAlgebra):
  with(plots):
> EgriAnaliz:=proc(e,f,F)
  local a,b,c,ev,P1,P2,P3,V1,V2,V3,
        J1,J2,J3,FV1,FV2,FV3,FV1j,FV2j,FV3j,
        i,j,subst1,subst2,subst3,
        Egri,Nok1,Nok2,Nok3,Vek1,Vek2,Vek3:
  a:=e[1]: b:=e[2]: c:=e[3]:
  print("Uzay egrisi e(t)"=[a,b,c]):
  print("f"=f):
  print("F"=F):
  print("--- i: Noktalari ---"):
  P1:=simplify(subs(t=-2,[a,b,c])):
  P2:=simplify(subs(t=0,[a,b,c])):
  P3:=simplify(subs(t=2,[a,b,c])):
  print("t=-2 P1"=P1):
  print("t=0  P2"=P2):
  print("t=2  P3"=P3):
  print("--- ii: Hiz Vektorleri ---"):
  ev:=[diff(a,t),diff(b,t),diff(c,t)]:
  print("egri turevi"=ev):
  V1:=simplify(subs(t=-2,ev)):
  V2:=simplify(subs(t=0,ev)):
  V3:=simplify(subs(t=2,ev)):
  print("t=-2 V1"=V1):
  print("t=0  V2"=V2):
  print("t=2  V3"=V3):
  print("--- iii: Grafik ---"):
  Egri:=spacecurve([a,b,c],t=-3..3,color=red,thickness=2):
  Nok1:=pointplot3d(P1,symbol=circle,color=blue,symbolsize=15):
  Nok2:=pointplot3d(P2,symbol=circle,color=green,symbolsize=15):
  Nok3:=pointplot3d(P3,symbol=circle,color=orange,symbolsize=15):
  Vek1:=spacecurve([P1[1]+s*V1[1],P1[2]+s*V1[2],P1[3]+s*V1[3]],
        s=0..1,color=blue,thickness=3):
  Vek2:=spacecurve([P2[1]+s*V2[1],P2[2]+s*V2[2],P2[3]+s*V2[3]],
        s=0..1,color=green,thickness=3):
  Vek3:=spacecurve([P3[1]+s*V3[1],P3[2]+s*V3[2],P3[3]+s*V3[3]],
        s=0..1,color=orange,thickness=3):
  print(display({Egri,Nok1,Nok2,Nok3,Vek1,Vek2,Vek3},axes=normal,
        scaling=constrained,orientation=[30,60]));
  print("--- iv: (F*)p(Vp) Hesabi ---"):
  subst1:={x[1]=P1[1],x[2]=P1[2],x[3]=P1[3]}:
  subst2:={x[1]=P2[1],x[2]=P2[2],x[3]=P2[3]}:
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subst3:={x[1]=P3[1],x[2]=P3[2],x[3]=P3[3]}:
FV1:=[V1[1]*diff(F[1],x[1])+V1[2]*diff(F[1],x[2])+V1[3]*diff(F[1
],x[3]),

V1[1]*diff(F[2],x[1])+V1[2]*diff(F[2],x[2])+V1[3]*diff(F[2],x[3]
),

V1[1]*diff(F[3],x[1])+V1[2]*diff(F[3],x[2])+V1[3]*diff(F[3],x[3]
)]:
FV1:=simplify(subs(subst1,FV1)):
print("P1 icin (F*)p(V1)"=FV1):
FV2:=[V2[1]*diff(F[1],x[1])+V2[2]*diff(F[1],x[2])+V2[3]*diff(F[1
],x[3]),

V2[1]*diff(F[2],x[1])+V2[2]*diff(F[2],x[2])+V2[3]*diff(F[2],x[3]
),

V2[1]*diff(F[3],x[1])+V2[2]*diff(F[3],x[2])+V2[3]*diff(F[3],x[3]
)]:
FV2:=simplify(subs(subst2,FV2)):
print("P2 icin (F*)p(V2)"=FV2):
FV3:=[V3[1]*diff(F[1],x[1])+V3[2]*diff(F[1],x[2])+V3[3]*diff(F[1
],x[3]),

V3[1]*diff(F[2],x[1])+V3[2]*diff(F[2],x[2])+V3[3]*diff(F[2],x[3]
),

V3[1]*diff(F[3],x[1])+V3[2]*diff(F[3],x[2])+V3[3]*diff(F[3],x[3]
)]:
FV3:=simplify(subs(subst3,FV3)):
print("P3 icin (F*)p(V3)"=FV3):
print("--- v: Jakobiyen Matrisleri ---"):
J1:=Matrix(3,3):
J2:=Matrix(3,3):
J3:=Matrix(3,3):
for i from 1 to 3 do
for j from 1 to 3 do
J1[i,j]:=simplify(subs(subst1,diff(F[i],x[j]))):
J2[i,j]:=simplify(subs(subst2,diff(F[i],x[j]))):
J3[i,j]:=simplify(subs(subst3,diff(F[i],x[j]))):
od:
od:
print("P1 Jakobiyen"): print(J1):
print("P2 Jakobiyen"): print(J2):
print("P3 Jakobiyen"): print(J3):
print("--- vi: Jakobiyen ile (F*)p(Vp) ---"):
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FV1j:=simplify(J1.Vector(V1)):
FV2j:=simplify(J2.Vector(V2)):
FV3j:=simplify(J3.Vector(V3)):
print("Jakobiyen ile P1 icin (F*)p(V1)"=FV1j):
print("Jakobiyen ile P2 icin (F*)p(V2)"=FV2j):
print("Jakobiyen ile P3 icin (F*)p(V3)"=FV3j):
end:
> e1:=[t,t^2,t^3]:
f1:=x[1]^2+x[2]^2+x[3]^2:
F1:=[x[1]*x[2],x[2]*x[3],x[1]*x[3]]:
EgriAnaliz(e1,f1,F1);
```

"Uzay egrisi e(t)" = $[t, t^2, t^3]$

"f" = $x_1^2 + x_2^2 + x_3^2$

"F" = $[x_1 x_2, x_2 x_3, x_1 x_3]$

"--- i: Noktalari ---"

"t=-2 P1" = [-2, 4, -8]

"t=0 P2" = [0, 0, 0]

"t=2 P3" = [2, 4, 8]

"--- ii: Hiz Vektorleri ---"

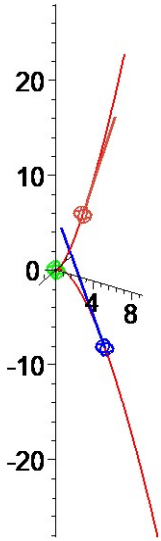
"egri turevi" = $[1, 2\,t, 3\,t^2]$

"t=-2 V1" = [1, -4, 12]

"t=0 V2" = [1, 0, 0]

"t=2 V3" = [1, 4, 12]

"--- iii: Grafik ---"



"--- iv: (F*)p(Vp) Hesabi ---"

"P1 icin (F*)p(V1)" = [12, 80, -32]

"P2 icin (F*)p(V2)" = [0, 0, 0]

"P3 icin (F*)p(V3)" = [12, 80, 32]

"--- v: Jakobiyen Matrisleri ---"

"P1 Jakobiyen"

$$\begin{bmatrix} 4 & -2 & 0 \\ 0 & -8 & 4 \\ -8 & 0 & -2 \end{bmatrix}$$

"P2 Jakobiyen"

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

"P3 Jakobiyen"

$$\begin{bmatrix} 4 & 2 & 0 \\ 0 & 8 & 4 \\ 8 & 0 & 2 \end{bmatrix}$$

"--- vi: Jakobiyen ile (F*)p(Vp) ---"

$$\text{"Jakobiyen ile P1 icin (F*)p(V1)"} = \begin{bmatrix} 12 \\ 80 \\ -32 \end{bmatrix}$$

$$\text{"Jakobiyen ile P2 icin (F*)p(V2)"} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{"Jakobiyen ile P3 icin (F*)p(V3)"} = \begin{bmatrix} 12 \\ 80 \\ 32 \end{bmatrix}$$

```
> e2:=[cos(t),sin(t),t]:
  f2:=x[1]^2+x[2]^2+x[3]:
  F2:=[x[1]^2,x[2]^2,x[3]^2]:
  EgriAnaliz(e2,f2,F2);
```

$$\text{"Uzay egrisi e(t)"} = [\cos(t), \sin(t), t]$$

$$\text{"f"} = x_1^2 + x_2^2 + x_3$$

$$\text{"F"} = [x_1^2, x_2^2, x_3^2]$$

"--- i: Noktalari ---"

$$\text{"t=-2 P1"} = [\cos(2), -\sin(2), -2]$$

$$\text{"t=0 P2"} = [1, 0, 0]$$

$$\text{"t=2 P3"} = [\cos(2), \sin(2), 2]$$

"--- ii: Hiz Vektorleri ---"

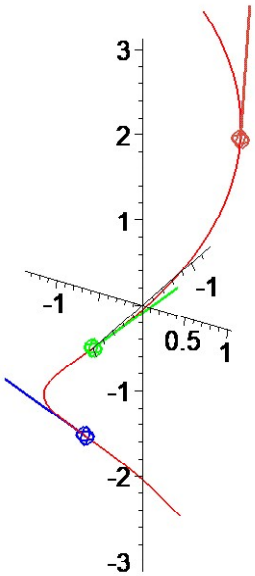
$$\text{"egri turevi"} = [-\sin(t), \cos(t), 1]$$

$$\text{"t=-2 V1"} = [\sin(2), \cos(2), 1]$$

$$\text{"t=0 V2"} = [0, 1, 1]$$

$$\text{"t=2 V3"} = [-\sin(2), \cos(2), 1]$$

"--- iii: Grafik ---"



"--- iv: (F*)p(Vp) Hesabi ---"

"P1 icin (F*)p(V1)" = $[2\sin(2)\cos(2), -2\sin(2)\cos(2), -4]$

"P2 icin (F*)p(V2)" = $[0, 0, 0]$

"P3 icin (F*)p(V3)" = $[-2\sin(2)\cos(2), 2\sin(2)\cos(2), 4]$

"--- v: Jakobiyen Matrisleri ---"

"P1 Jakobiyen"

$$\begin{bmatrix} 2\cos(2) & 0 & 0 \\ 0 & -2\sin(2) & 0 \\ 0 & 0 & -4 \end{bmatrix}$$

"P2 Jakobiyen"

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

"P3 Jakobiyen"

$$\begin{bmatrix} 2\cos(2) & 0 & 0 \\ 0 & 2\sin(2) & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

"--- vi: Jakobiyen ile (F*)p(Vp) ---"

$$\text{"Jakobiyen ile P1 icin (F*)p(V1)"} = \begin{bmatrix} 2\sin(2)\cos(2) \\ -2\sin(2)\cos(2) \\ -4 \end{bmatrix}$$

$$\text{"Jakobiyen ile P2 icin (F*)p(V2)"} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{"Jakobiyen ile P3 icin (F*)p(V3)"} = \begin{bmatrix} -2\sin(2)\cos(2) \\ 2\sin(2)\cos(2) \\ 4 \end{bmatrix}$$

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